SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I J. Kirthiga / 192421265 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J. Kirthiga

19/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. IPV4 NETWORK CONFIGURATION DEBUGGING

The IP address 192.168.10.65/26 belongs to the subnet 192.168.10.64/26. The network address is 192.168.10.64, the broadcast address is 192.168.10.127, and the valid host range is 192.168.10.65–192.168.10.126. The IP address is valid, but the gateway 192.168.10.1 is in a different subnet. Changing the gateway to 192.168.10.126 fixes the issue. A /26 subnet has 62 usable host addresses.

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

Answer

1. Subnet of the IP Address
   - 192.168.10.65 belongs to 192.168.10.64/26
2. Network Details
   - Network Address: 192.168.10.64
   - Broadcast Address: 192.168.10.127
   - Valid Host Range: 192.168.10.65 – 192.168.10.126
3. Verify IP and Gateway
   - IP Address (192.168.10.65) → Valid
   - Default Gateway (192.168.10.1) → Invalid (belongs to a different subnet)
4. Configuration Error
   - The default gateway is not in the same subnet as the workstation.
5. Optimized Configuration
   - IP Address: 192.168.10.65
   - Default Gateway: 192.168.10.126 (or any valid host in the same subnet)
6. Usable Host Addresses
   - $2^6 - 2 = 62$
   - Usable Hosts = 62

2. IPV6 CONFIGURATION DEBUGGING

Answer

Both IPv6 addresses expand correctly and use the prefix 2001:db8:0:0::/64, so they are in the same subnet. There is no addressing mismatch, and the issue is likely due to routing or interface settings. A /64 subnet provides 2^{64} host addresses

1. Expanded Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix

- Device A: 2001:db8:0:0::/64
- Device B: 2001:db8:0:0::/64

3. Same Subnet?

- Yes, both are in the same /64 subnet.

4. Addressing Issue

- No prefix mismatch. Communication failure is likely due to another configuration issue (routing, firewall, or interface).

5. Optimized Configuration

- Device A: 2001:db8:0:0::10/64
- Device B: 2001:db8:0:0::20/64

6. Available Host Addresses

- 2^{64} addresses (approximately 18.4 quintillion)

3.IPV4 SUBNET ANALYSIS AND VLSM

The /26, /27, and /28 subnets provide 62, 30, and 14 usable hosts respectively. Fixed subnet allocation can waste addresses. Using VLSM allocates subnet sizes based on department needs, improving address usage.

Subnet	Network Address	Host Range	Broadcast	Usable Hosts
/26	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63	62
/27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95	30
/28	192.168.1.96	192.168.1.97 – 192.168.1.110	192.168.1.111	14

Unused Addresses

- /26 → 64 total (62 usable)
- /27 → 32 total (30 usable)
- /28 → 16 total (14 usable)

Inefficient Allocation

- Large departments may need more than /28.
- Small departments waste addresses with /26.

Addressing Mistake

- Fixed-size subnet allocation causes inefficient IP usage.

Optimized VLSM

- Large department → /26
- Medium department → /27
- Small department → /28
- Allocate larger subnets first to improve utilization

4. IPv6 Configuration Debugging

Both devices belong to the 2001:db8:0:0::/64 subnet. Since the prefixes match, the addressing is correct. The communication problem is likely caused by routing or firewall issues. A /64 subnet supports 2^{64} addresses.

1. Expanded Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix

- Device A: 2001:db8:0:0::/64
- Device B: 2001:db8:0:0::/64

3. Same Subnet?

- Yes

4. Addressing Error

- No IPv6 addressing mismatch. Check routing, interface status, or firewall.

5. Optimized Addresses

- Device A → 2001:db8:0:0::10/64
- Device B → 2001:db8:0:0::20/64

6. Available Hosts

- 2^{64} host addresses.

5. ROUTING TABLE DEBUGGING

The destination 192.168.2.100 matches the route 192.168.2.0/24, so the next hop is 10.0.0.3. The routing entry is correct. If no route matches, the router forwards the packet using the default route (10.0.0.1).

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Answer

1. Longest-Prefix Match
    - Destination 192.168.2.100 matches 192.168.2.0/24
2. Next-Hop Router
    - 10.0.0.3
3. Destination Verification
    - 192.168.2.100 belongs to 192.168.2.0/24
4. Routing Error
    - The routing table is correct. If packets fail, verify the next-hop router, interface status, or destination network.
5. Optimized Routing
    - Ensure the next-hop 10.0.0.3 is reachable and remove any unnecessary routes to improve efficiency.
6. Default Route
    - If no matching route exists, packets are forwarded to 10.0.0.1.